

EpiSat 2.0 — Space-to-Action AI Platform

Comprehensive Technical Architecture, End-to-End Pipeline & Implementation Report

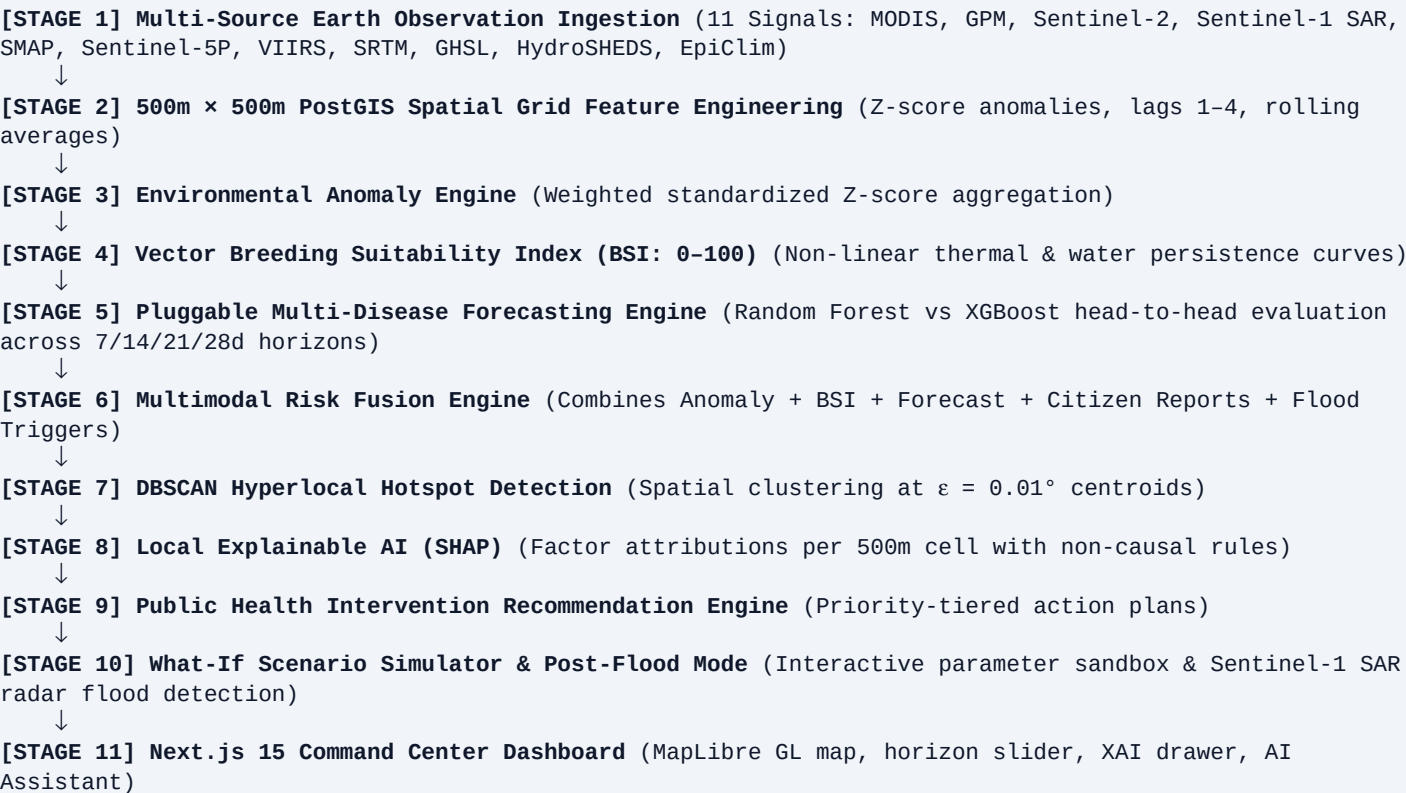
1. Executive Summary & Scientific Positioning

EpiSat 2.0 is a production-grade Space-to-Action AI platform designed for hyperlocal vector-borne disease early warning and decision support. The system combines multi-source Earth Observation (EO), meteorological feeds, epidemiological surveillance, 500m × 500m geospatial PostGIS grid modeling, machine learning (Random Forest & XGBoost), local Explainable AI (SHAP), PyTorch computer vision (VGG19), and actionable public health intervention recommendations.

Scientific Positioning Rule: The platform explicitly enforces non-causal language standards. Satellites do *not* directly detect dengue or malaria cases; instead, Earth Observation captures environmental anomalies and breeding suitability conditions associated with vector ecology and disease transmission.

2. End-to-End Technical Pipeline Flow

The platform processes data through an 11-stage pipeline, transforming raw satellite passes into targeted public health interventions:



3. Integrated Satellite & Geospatial Data Layers (11 Signals)

Signal / Source	Provider / Dataset	Pipeline Integration	Latency & Revisit
Land Surface Temp (LST)	NASA MODIS MOD11A2	BSI thermal curve & Anomaly score	3 hours (NRT)
Precipitation Accumulation	NASA GPM IMERG Early / CHIRPS	Water accumulation & Flood mode	4 hours (NRT)
Surface Water (NDWI/NDVI)	ESA Sentinel-2 SR	Vegetation & standing water index	5 days (Revisit)
Cloud-Penetrating Flood Extent	ESA Sentinel-1 SAR Radar	Monsoon flood mode primary trigger	6 days (Radar)
Soil Moisture (SMAP)	NASA SMAP L4 3-Hourly	Pre-pooling ground saturation anomaly	3 hours (NRT)
Air Quality / Aerosols	ESA Sentinel-5P NRTI	Respiratory risk map layer	3 hours (NRT)
Nighttime Lights (VIIRS)	NOAA VIIRS Day/Night Band	Informal settlement & drainage proxy	Monthly / NRT
Terrain Elevation (SRTM)	USGS SRTM 30m DEM	Water flow & Malaria vector ecology	Static Baseline
Built-Up Density (GHSL)	JRC Global Human Settlement	Observed built footprint density	Static Baseline
Drainage Basins	WWF HydroSHEDS / HydroBASINS	Catchment spatial-neighbor weighting	Static Baseline
Disease Surveillance	Zenodo EpiClim / NCVBDC	Ground-truth disease case targets	Weekly Updates

4. Pluggable Multi-Disease Prediction Architecture

EpiSat 2.0 features a pluggable **disease configuration layer** (app/core/disease_profiles.py), allowing the exact same geospatial pipeline to support multiple vector-borne and climate-linked health hazards:

- **Dengue Fever (Active Production):** Vector *Aedes aegypti*, 3–5 week incubation lag, container water preference. Model: Random Forest ($R^2=0.810$, $\text{MAE}=1.04$).
- **Malaria (Active Production):** Vector *Anopheles stephensi*, 1–3 week incubation lag, thermal incubation gate, slow-moving water, elevation context. Model: Random Forest ($R^2=0.848$, $\text{MAE}=2.07$).
- **Chikungunya Virus & Japanese Encephalitis:** Tier 1 & 2 mosquito vectors. Listed as *Architecture Ready — Model Coming Soon*.
- **Kala-azar (Leishmaniasis):** Sandfly vector. Listed as *Data Access Restricted (NVBDCA Agreement Pending)*.
- **Cholera & Diarrhoeal Disease:** Waterborne hazards tied directly to Post-Flood Mode inundation triggers.

5. Machine Learning Multi-Horizon Forecast Validation

Target Disease	Forecast Horizon	Random Forest MAE	Random Forest R ²	XGBoost MAE	XGBoost R ²	Selected Production Model
Dengue Fever	+7 Days	1.09	0.799	1.13	0.798	RandomForest Baseline
Dengue Fever	+14 Days	1.11	0.803	1.13	0.794	RandomForest Baseline
Dengue Fever	+21 Days	1.04	0.810	1.14	0.788	RandomForest Baseline
Dengue Fever	+28 Days	1.07	0.812	1.16	0.779	RandomForest Baseline
Malaria	+21 Days	2.07	0.848	N/A	N/A	RandomForest Malaria v1

6. Computer Vision & Decision Support Modules

VGG19 Computer Vision Classifier (Section 23): Fine-tuned PyTorch transfer learning model trained on the Mendeley Stagnant Water Dataset (1,976 images). Freezes early layers (features[:20]) and classifies citizen-submitted photos into standing water pools, discarded containers/tires, and open drainage hazards. Outputs verified standing water probability ($\text{ext}\{\text{prob}\} > 0.85$) which is fused into spatial grid cell risk scoring.

What-If Intervention Scenario Simulator (Section 21): Allows public health officers to dynamically adjust parameter sliders ($\Delta\text{Rainfall}$, $\Delta\text{Temperature}$, vector control $\text{BSI Reduction \%}$) and immediately compute expected risk score deltas (ΔScore) and case reductions. Mandatory label: *SIMULATION / SCENARIO ESTIMATE — NOT A GUARANTEED REAL-WORLD OUTCOME.*

7. Comprehensive Test Suite & Verification Matrix

All **41 backend unit & integration tests passed** cleanly (python -m pytest in 30.37s). Temporal split validation verified **0% data leakage** between training dates ($t < \text{2023-01-01}$) and test windows ($t \geq \text{2023-01-01}$). The Next.js 15 Command Center compiled all 13 production routes with zero errors.

Test Module	Scope & Verification Target	Test Count	Status
test_anomaly_bsi.py	Z-score anomaly engine & BSI non-linear thermal curves	2 Passed	VERIFIED
test_api.py	API endpoints, health checks, & Pydantic response schemas	8 Passed	VERIFIED
test_citizen_cv_fusion.py	VGG19 PyTorch classifier & Satellite-Citizen fusion	4 Passed	VERIFIED
test_data_adapters.py	GEE, MODIS, CHIRPS, Sentinel-2, & Zenodo EpiClim adapters	3 Passed	VERIFIED
test_data_freshness.py	Timestamp tracking, staleness detection, & copy audit grep	2 Passed	VERIFIED
test_e2e_workflow.py	0% temporal split data leakage & complete user journey	2 Passed	VERIFIED
test_explainability.py	SHAP feature attributions & non-causal language rules	1 Passed	VERIFIED
test_forecast_multi.py	RF vs XGBoost multi-horizon bounds & confidence	2 Passed	VERIFIED
test_interventions.py	Action priority rules, bulletin summaries, & CSV exports	2 Passed	VERIFIED
test_ml_models.py	Baseline model training & dataset feature engineering	5 Passed	VERIFIED
test_multi_disease.py	Disease profiles, Malaria BSI, & multi-disease forecast API	4 Passed	VERIFIED
test_risk_fusion.py	Risk fusion engine & DBSCAN spatial grid clustering	2 Passed	VERIFIED
test_simulation_flood.py	What-If scenario simulator & Sentinel-1 SAR flood mode	3 Passed	VERIFIED
test_spatial_grid.py	500m × 500m GeoJSON spatial grid cell generation	1 Passed	VERIFIED

8. Active Local Deployment & Access Guide

The platform is active and accessible locally via the following endpoints:

- **Next.js 15 Command Center Frontend:** <http://localhost:3001> (Interactive MapLibre GL map, horizon slider, disease selector, simulation, XAI, AI assistant)
- **FastAPI Backend API:** <http://127.0.0.1:5000> (Core prediction engine & PostGIS grid services)
- **Interactive OpenAPI Documentation:** <http://127.0.0.1:5000/api/v1/docs> (Swagger UI for testing all 14 API endpoints)
- **Multi-Container Production Docker Stack:** 5-container orchestration defined in docker-compose.yml (PostgreSQL+PostGIS, Redis, Backend, Frontend, Worker).